CLAIMS

What is claimed is:

1. A method for secure communication between a sender and a receiver, comprising the steps of:

    - at the sender, generating a first secret key, K1, which is not transmitted to the receiver;
    - at the sender, encrypting a message M using K1 to produce an encrypted message;
    - transmitting the encrypted message from the sender to the receiver;
    - transmitting a prefix data block P from the sender to the receiver;
    - transmitting a suffix data block S from the sender to the receiver;
    - at the sender, deriving a second key, K2, from K1 and the combination of P, the encrypted message, and S;
    - transmitting K2 from the sender to the receiver; and
    - at the receiver, decrypting the combination of P, the encrypted message, and S using K2.

2. The method of claim 1, wherein the encrypted message, the prefix P, the suffix S, and the second key K2 are transmitted at different points in time.

3. The method of claim 1, wherein at least two of the transmissions from the group consisting of the encrypted message, the prefix P, the suffix S, and the second key K2, are performed over different communication channels.

4. The method of claim 1, wherein the step of deriving K2 comprises computing a cryptographic hash of the concatenation of P, the encrypted message, and S, and using the hash and K1 as inputs to a key derivation function.

5. A system for secure communication, comprising:

    - a sender device configured to:
        - generate a non-transmitted first key K1;
        - encrypt a message M with K1;
        - transmit the encrypted M, a prefix P, and a suffix S in separate transmissions;
        - derive a second key K2 from K1 and the combination of P, M, and S; and
        - transmit K2;
    - a receiver device configured to:
        - receive the encrypted M, P, S, and K2;
        - assemble the received P, M, and S; and
        - decrypt the assembled data using K2.

6. The system of claim 5, wherein the sender device is configured to perform the transmissions asynchronously.